

Transfer Learning Models

- Generally we trained the model on the given data
- ML or DL, we save the model in one format
- pickle or Job lib For Machine learning
- Deep learning tf.save or using keras also
- In the deep learning model saving means , saving the weight file
- You develop the model with sequential order
- This is called model configuration
- Model config =====> Model architecture
- Model layers means ===== each layer
- Model summary ===== each layer parameters
- Model weights means ===== final weights
- we developed and we saved: Custom model
- I given that saved model to the client
- Client will use that model : Transfer learning
- The models which are using as Transfer learning called as Pretrained models
- Pre trained means already trained model

CNN: Object detection

LeNET ===== Pretrained models

AlexNet

VGGnet

VGG16 – 32

MobileNet

Google Net

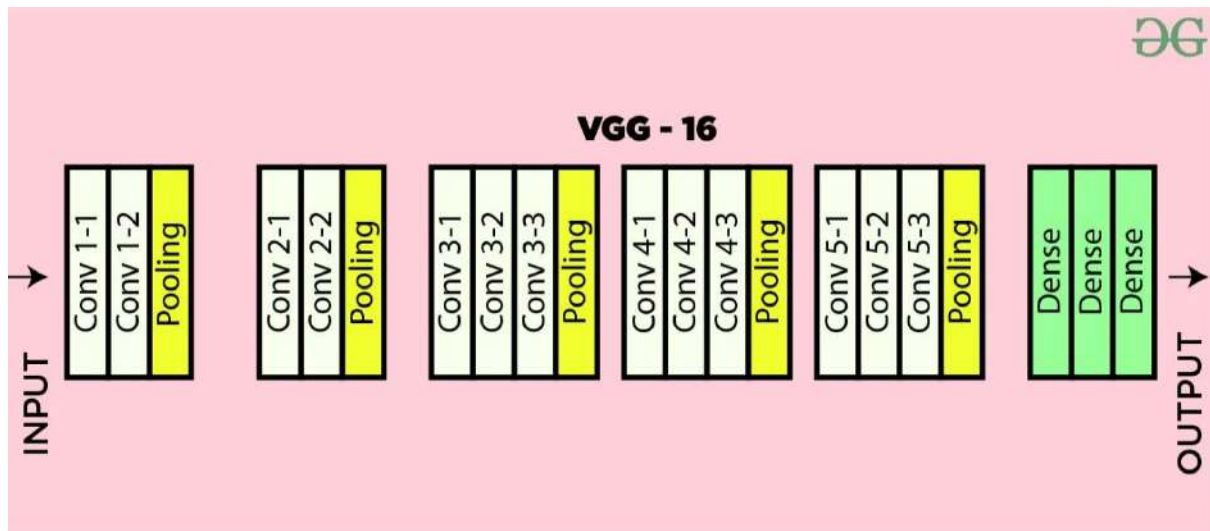
ResNet

FastRCNN

MaskRCNN

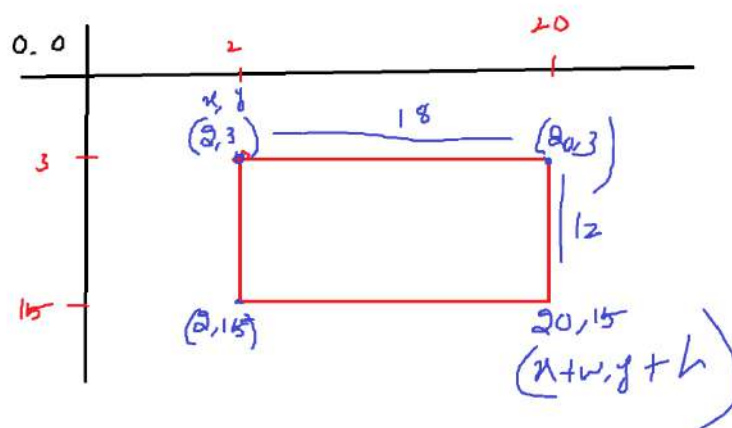
FasterRCNN

YOLO V8

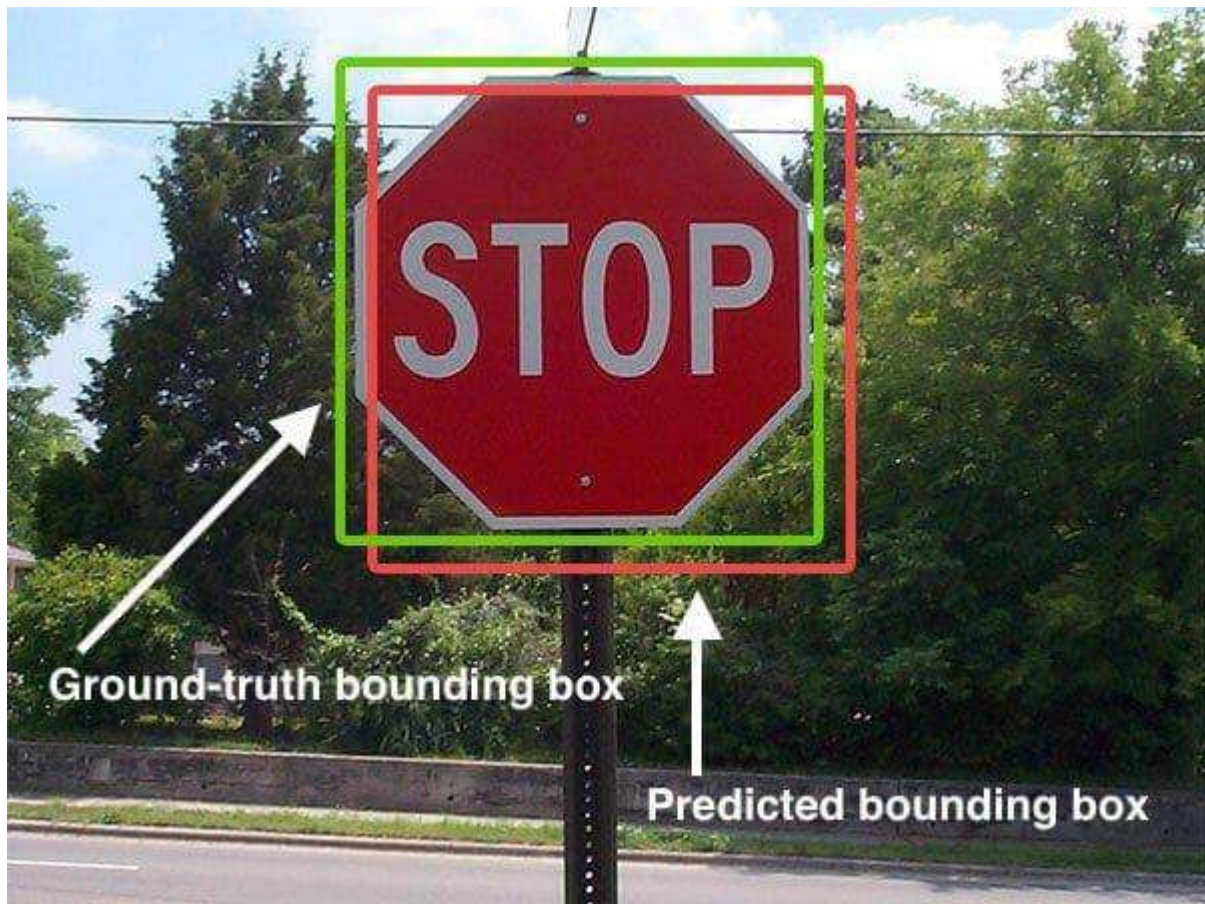


- Consider MobileNet trained on 1.3 Million images
- Identify the objects around 1000 classes
- Object identification model
- we have two types of use cases
 - Object identification : recognize the object and classify also
 - object Detection : Extract the location also
- Object detection is used to extract the coordinates of object is called as Bounding box
- Bounding box means a rectangular box on the images , rectangular box will give $(x, y, x + w, y + h)$

top left corner = (x, y) bottom right = $(x + w, y + h)$



Inter section of Union (IOU)

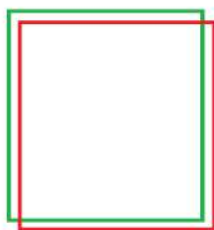


Actual boxes : Ground truth

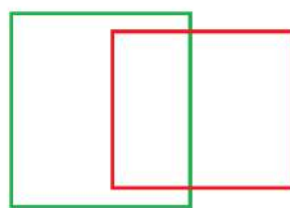
Ground truth bbox vs Prediction bbox



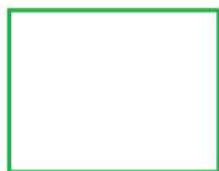
a



b



c

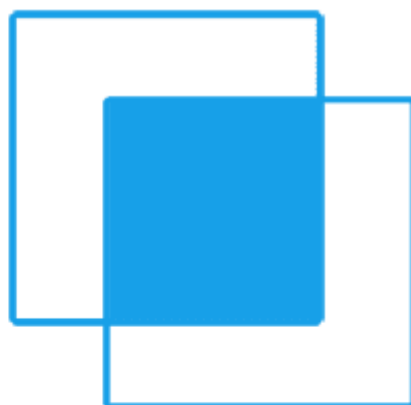


d



$$\begin{aligned} GT &= A & P &= B \\ \frac{A \cap B}{A \cup B} \end{aligned}$$

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



Fine Tuning

Consider MobileNet trained on 1.3 Million images

- *Identify the objects around 1000 classes*
- *Imagine that you want identify your college ID card*
- *This model is not trained on your clg id card*
- *so It will not able to identify the clg id card*
- *You need to collect some images of your clg id card*
- *You need to create a dataset : around 100 images*
- *Again you need to train the MobileNet model*
- *Mobile net already has 1000 classes + 1 classes*
- *according to this again weights will be updated*
- *Fine Tuning*